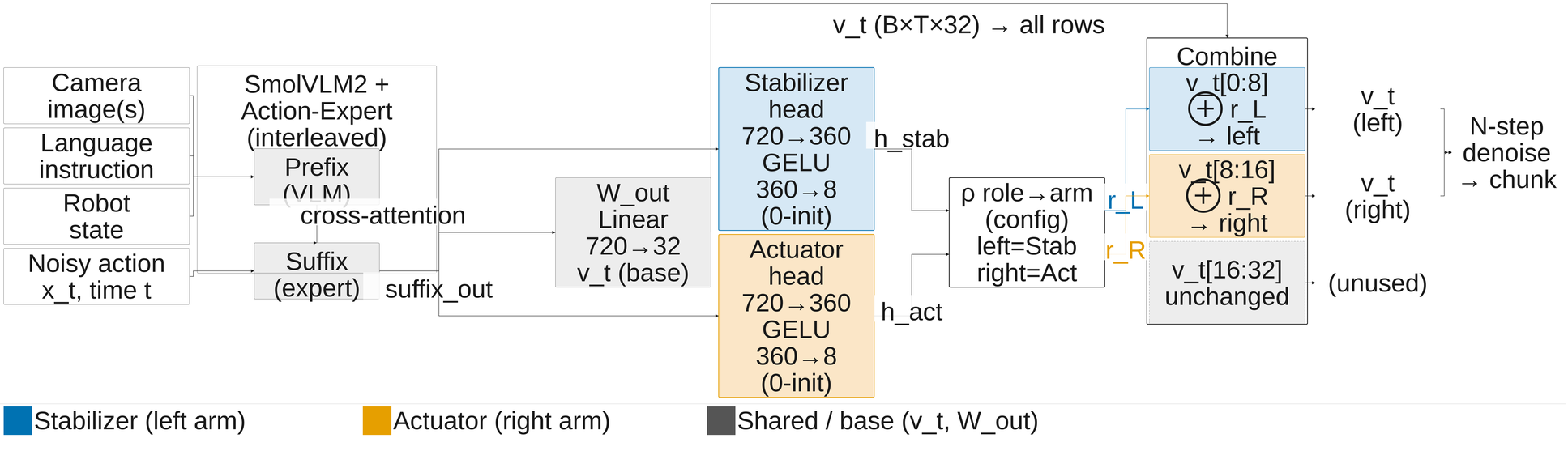
Camera image(s)
Language instruction
Robot state
Noisy action x_t, time t
SmolVLM2 + Action-Expert (interleaved)
Prefix (VLM)
cross-attention
Suffix (expert)
suffix_out
W_out Linear 720→32 v_t (base)
v_t (B×T×32) → all rows
Stabilizer head 720→360 GELU 360→8 (0-init)
Actuator head 720→360 GELU 360→8 (0-init)
h_stab
h_act
ρ role→arm (config) left=Stab right=Act
r_L
r_R
Combine
v_t[0:8] ⊕ r_L → left
v_t[8:16] ⊕ r_R → right
v_t[16:32] unchanged
v_t (left)
v_t (right)
(unused)
N-step denoise → chunk
Stabilizer (left arm)
Actuator (right arm)
Shared / base (v_t, W_out)